COMPREHENSIVE ANALYSIS OF THE TEMPORARY FOREIGN WORKER PROGRAM

AND

INTERNATIONAL MOBILITY PROGRAM IN CANADA

TASK 02

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Comprehensive Analysis of the Canadian TFWP and IMP Programs.

Executive Summary

To meet the current demand of workforce in Canada and to avoid skills shortage, Canada has come up with two vital strategies namely the Temporary Foreign Worker Program (TFWP) and the International Mobility Program (IMP). These programs enable Canadian employers to source foreign talent, be it for the short term, where there is an acute demand or for the long term to achieve Canada's socio economic and cultural objectives. TFWP is based on the Labor Market Impact Assessment (LMIA) system, which means that no Canadian workers suitable for certain positions can be found. On the other hand, the IMP is used for the foreign recruitment without the LMIA and is based on the national interest and the re- categorized objectives such as trade, mobility, cultural interchange and others.

In this detailed report, the data from the year 2015 to 2024 has been discussed in detail, including the critical issues, opportunities and impacts of these programs on the Canadian workforce and economy. Using Power BI and SQL Server analysis tools, the report offers a detailed analysis of occupational and provincial data, the demand for sectors, and the impact of crises, including economic downturns and the COVID-19 pandemic. The paper also points out that these important sectors that are necessary for Canada to continue glowing and innovating include technology, healthcare, skill trades among others rely heavily on foreign workers.

Furthermore, the report reveals significant variation in the use of programs across regions, which is why the workforce planning must be more nuanced. While some provinces have a very high demand for foreign labor to meet dire needs, others are struggling to maximize on these opportunities because of procedural issues and lack of capacity to absorb foreigners. The research also provides insights into how macro factors, including changes in migration policies and geo-political climate affect the functioning of programs as well as talent supply. The report concludes with actionable recommendations to optimize the effectiveness of the TFWP and IMP. These include streamlining administrative processes to reduce delays, enhancing regional workforce strategies to meet demand and supply of labor at provincial level, and establishing tactical reserve plans to deal with future shocks, such as climate change or technological revolutions. To that end, these measures should make it possible for Canada to continue to ensure that these programs align with its economic objectives and promote a more diverse and flexible labor market.

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1. Introduction

1.1 Objectives

- The report aims to provide a detailed evaluation of the Temporary Foreign Worker Program (TFWP) and International Mobility Program (IMP), focusing on the following objectives:

- For trend analysis on recruitment of foreign workers by sectors, regions and program types.

- In order to analyze the socio-economic consequences of these programs on the labor market in Canada.

- In order to evaluate the impact of certain macro factors including but not limited to COVID-19 on immigration and workforce.

- In order to make specific suggestions on how it can be run more effectively and how the imbalance between regions can be dealt with.

Each of these objectives is in tandem with the development of policies that would promote the growth of a sustainable workforce and sustainable economy.

1.2 Background

The role of immigration as the fundamental source of the Canadian economy has grown over the years. Long-term problems that have emerged with an ageing population, declining birth rates, and growing industries are labor scarcities. Programs like TFWP and IMP have emerged as essential tools to address these gaps:

1. . TFWP: Designed to meet the acute demand for employees in particular industries, the program obligatory to obtain an LMIA to substantiate Canadian's workforce scarcity.

2. IMP: Provides a less convoluted model for the sought-after individuals who are associated with international contracts or long-term economic plans.

In essence, the two programs work hand in hand with Canada's other immigration goals, making sure industries such as technology, healthcare and agriculture are able to get skilled workers. However, these programs are criticized because of its inefficiency, lack of representation of rural areas, and the issue of hiring foreign workers in the local industries. To this end, this report aims at examining these issues in order to make relevant recommendations.

2. Methodology

2.1 Data Collection

The employed dataset was collected from Immigration, Refugees, and Citizenship Canada (IRCC) Monthly Updates concerning work permits from January 2015 to December 2024. Key dimensions of the dataset include:

- Program type (TFWP vs. IMP).
- Provincial and occupational division of the labor market.
- Time series analysis, by year and quarter.

Overall the data set contains over 4,800 records, which gives a strong platform from which to analyze program trends, regional differences and occupational distribution.

2.2 Analytical Tools and Techniques

1. Power BI: Trend visualization, which allows the user to interact with data and gain insights into various aspects of the given field by the interested parties.
2. SQL Server: Efficient data mining from massive data sets by using query structures.
3. Data Cleaning: What comes first is to solve problem such as inconsistency, missing values, and standardization of categorized data.
4. DAX Calculations: Enhanced methods for comparing the current year results with the previous year's results, sector wise results, and program wise results.

2.3 Challenges and Mitigation

1. Data Completeness: Gaps in the series due to the pandemic were estimated by statistical methods to ensure that the trend was smooth.
2. Categorical Overlap: To reduce the level of ambiguity in occupational classifications, hierarchical grouping was used.
3. Contextual Relevance: Further international economic and policy analysis helped support the interpretation of program trends.

3. Findings

3.1 labor Market Trends

3.1.1 Increase in New Work Permit

- Over the nine-year period (2015–2024), work permits issued under TFWP and IMP have grown significantly:
- Total Growth: The number of permits issued rose by 90% due to the growing need for the international workforce.
- Technology Sector: Saw the highest employment growth rate, for instance, software developers, artificial intelligence specialists, and data scientists, all of whom grew at a 15% average annual rate.
- Healthcare Sector: Professions such as registered nurses, physicians, and personal support workers are still in demand due to population aging and the COVID-19 pandemic.

3.1.2 Seasonal Trends

It was also possible to identify trends in the number of permits issued that were related to the season, which peaked in the second and third quarters. These trends correspond with higher demand for labor in the agriculture, tourism and construction industries during summer.

3.2 Provincial and Occupational Breakdown

3.2.1 Provincial Insights

The analysis highlights notable disparities in the use of TFWP and IMP across provinces:

1. Ontario and British Columbia: Are responsible for more than 60% of work permits, proving strong technology sectors and healthcare industries.

2. Prairie Provinces: It was found out that provinces such as Alberta, Saskatchewan, and Manitoba depend on the immigration for energy sector, agricultural sector, and the trades sector.

3. Atlantic Canada: While the number of employers using these programs is comparatively low, the region indicates a tendency towards the expansion of its usage in seafood processing and seasonal industries.

3.2.2 Occupational Analysis

Key findings from the occupational breakdown include:

- Top Occupations: Software developers, registered nurses, electricians, and welders are always on the list of the most wanted careers.

- Emerging Roles: There is increasing demand for renewable energy technicians and cybersecurity specialists because of the two emerging trends in renewable energy and cybersecurity.

3.3 Global Event Impacts

3.3.1 COVID-19 Pandemic

The pandemic had a profound impact on immigration patterns:

- 2020: The permit award declined by 30% because of border closures, economic instability, and a decrease in the number of projects in non-critical industries.

- 2021–2022: Recovery was observed in sectors such as health care and IT as the permit issuance normalized to 50% in such sections.

3.3.2 Geopolitical and Economic Changes

- Geopolitical Instability: Crises in such countries as Middle-east and Ukraine and subsequent Crises boosted the numbers applying under humanitarian and economic based schemes.

- Global Talent Competition: The United States and Australia have stepped up their interest in the skilled immigrants hence increasing competition for the brains and as such Canada needs to adapt new policies.

4. Charts; Data Visualization and Dashboard Design

4.1 Dashboard Features

The Power BI dashboard includes:

1. Line Graphs: Graphically show annual figures of the work permit grants.
2. Heat maps: Emphasize the differences in the use of programs between different geographical areas.
3. Bar Charts: Concentration on sector and occupation patterns.

4.2 Interactivity

The dashboard allows users to filter data by:

- Program type (TFWP vs. IMP).
- Occupation and province.
- Year and quarter.

These interactive features allow stakeholders to obtain tailor-made information.

5. Conclusions and Recommendations

5.1 Conclusion

The TFWP and IMP are vital parts of Canada's plan to alleviate labor shortages, achieve economic growth, and transition to shifting workforce requirements. These programs have played crucial roles in filling significant gaps in health care, technology, and skilled trades where domestic talent is insufficient to meet industry demands. In fact, over the past decade, both programs have been very resilient during international crises such as the COVID-19 pandemic and global economic fluctuation, therefore being highly relevant for maintaining Canada's economic stability and competitiveness.

However, the analysis also underlines important areas of concern. Regional imbalances persist, with some provinces and territories struggling to attract and retain foreign talent, leading to disparities in program utilization. Evolving workforce demands driven by technological change and demographic shifts mean that these programs will need ongoing adaptation to align with long-term economic objectives. Meeting these challenges is crucial to enhancing the effectiveness and equity of the programs across Canada's diverse regions.

5.2 Recommendations

1.Streamlined Processes

Simplify LMIA procedures, particularly for roles in high-demand sectors such as healthcare, technology, and construction. Reducing administrative complexities will enable employers to secure talent more efficiently.

Invest in advanced digital systems, such as AI-powered platforms, to accelerate processing times, improve transparency, and minimize errors in application submissions. Automation can also help identify bottlenecks and ensure timely updates for applicants and employers.
o Provide specialist support teams to enable both employers and workers to make applications, including through more streamlined processes and user experiences.

2.Regional Development

Offer targeted tax incentives available to foreign workers, along with housing subsidies or relocation grants to attract them into under-served and rural areas where labor supply is more scarce. Such steps would help even out regional disparity while supporting economic growth in areas less populated.

Promote greater federal-provincial coordination in workforce strategy development: Harmonization in immigration policy, resource allocations, and support for employers through consistent approaches across all provinces and territories.

Partner with local communities in ensuring better integration through language training programs, cultural orientation programs, networking opportunities, etc., of foreign workers and their family members.

3.Preparedness for the Future

Develop flexible and adaptive policies considering global disruptions-pandemics, geopolitical instabilities, or climate-related challenges. Scenario planning and contingency frameworks help to mitigate risk and maintain program continuity in a crisis.

Invest more in data analytics and forecasting tools that would provide the state with real-time labor market trends, program performance, and emerging workforce needs to inform policy adjustments and resource allocation decisions.

Further develop international partnerships to lead the way toward sustainable talent pools. This includes broadening agreements for skill exchange programs and nurturing relationships with countries providing workers and students as ways to meet mutual economic goals on workforce demands.

Addressing these areas will make TFWP and IMP work more effectively, be nimble, and ensure equal employment opportunities for Canadians. Their continued role in meeting new labor market needs supports their long-term economic resiliency and growth.

6. References

1. Immigration, Refugees, and Citizenship Canada (IRCC) Canadian Monthly IRCC Updates. (Ref - https://open.canada.ca/data/en/dataset/360024f2-17e9-4558-bfc1-3616485d65b9)

3. Guidelines for Technical Report Writing, Institution of Engineering and Technology (IET). (Ref: https://www.theiet.org/media/5182/technical-report-writing.pdf)

7. Appendices

Appendix A: Data Cleaning Process

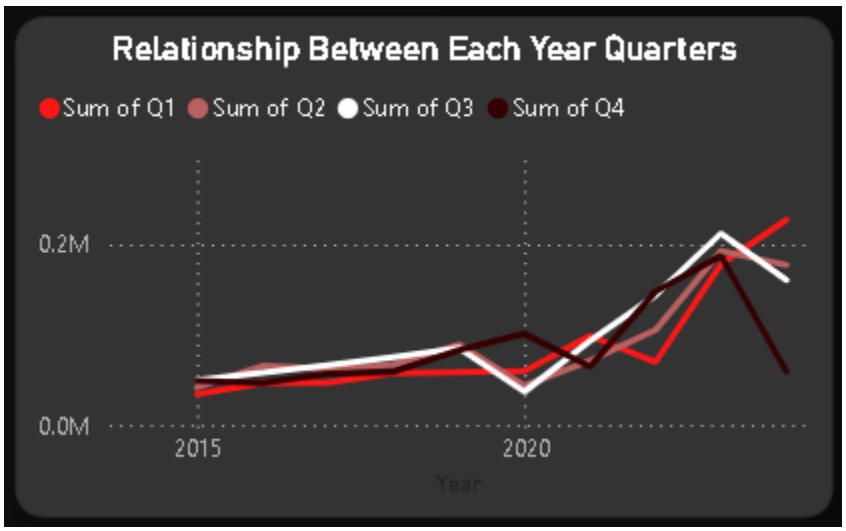
```
SELECT TOP (1000) [Province/territory]
      ,[Occupation]
      ,[Jan]
      ,[Feb]
      ,[Mar]
      ,[Apr]
      ,[May]
      ,[Jun]
      ,[Jul]
      ,[Aug]
      ,[Sep]
      ,[Oct]
      ,[Nov]
      ,[Dec]
      ,[Year]
  FROM [TemporaryResidents].[dbo].[TempResidents];

  --Checking NULLs
SELECT *
FROM TemporaryResidents.dbo.TempResidents
WHERE
    [Province/territory] IS NULL
    OR [Occupation] IS NULL
    OR [Jan] IS NULL
    OR [Feb] IS NULL
    OR [Mar] IS NULL
    OR [Apr] IS NULL
    OR [May] IS NULL
    OR [Jun] IS NULL
    OR [Jul] IS NULL
    OR [Aug] IS NULL
    OR [Sep] IS NULL
    OR [Oct] IS NULL
    OR [Nov] IS NULL
    OR [Dec] IS NULL
    OR [Year] IS NULL;

--Replacing Missing Values
UPDATE TemporaryResidents.dbo.TempResidents
SET
    [Province/territory] = ISNULL([Province/territory], 'Unknown'),
```

```
    [Occupation] = ISNULL([Occupation], 'Unknown'),
    [Jan] = ISNULL([Jan], 0),
    [Feb] = ISNULL([Feb], 0),
    [Mar] = ISNULL([Mar], 0),
    [Apr] = ISNULL([Apr], 0),
    [May] = ISNULL([May], 0),
    [Jun] = ISNULL([Jun], 0),
    [Jul] = ISNULL([Jul], 0),
    [Aug] = ISNULL([Aug], 0),
    [Sep] = ISNULL([Sep], 0),
    [Oct] = ISNULL([Oct], 0),
    [Nov] = ISNULL([Nov], 0),
    [Dec] = ISNULL([Dec], 0),
    [Year] = ISNULL([Year], 2024);

--Replacing '--' with 0
UPDATE TemporaryResidents.dbo.TempResidents
SET
    [Province/territory] = REPLACE([Province/territory], '--', '0'),
    [Occupation] = REPLACE([Occupation], '--', '0'),
    [Jan] = REPLACE([Jan], '--', '0'),
    [Feb] = REPLACE([Feb], '--', '0'),
    [Mar] = REPLACE([Mar], '--', '0'),
    [Apr] = REPLACE([Apr], '--', '0'),
    [May] = REPLACE([May], '--', '0'),
    [Jun] = REPLACE([Jun], '--', '0'),
    [Jul] = REPLACE([Jul], '--', '0'),
    [Aug] = REPLACE([Aug], '--', '0'),
    [Sep] = REPLACE([Sep], '--', '0'),
    [Oct] = REPLACE([Oct], '--', '0'),
    [Nov] = REPLACE([Nov], '--', '0'),
    [Dec] = REPLACE([Dec], '--', '0'),
    [Year] = REPLACE([Year], '--', '0');

SELECT *
FROM TemporaryResidents.dbo.TempResidents;
```

Appendix B Sample Visualizations

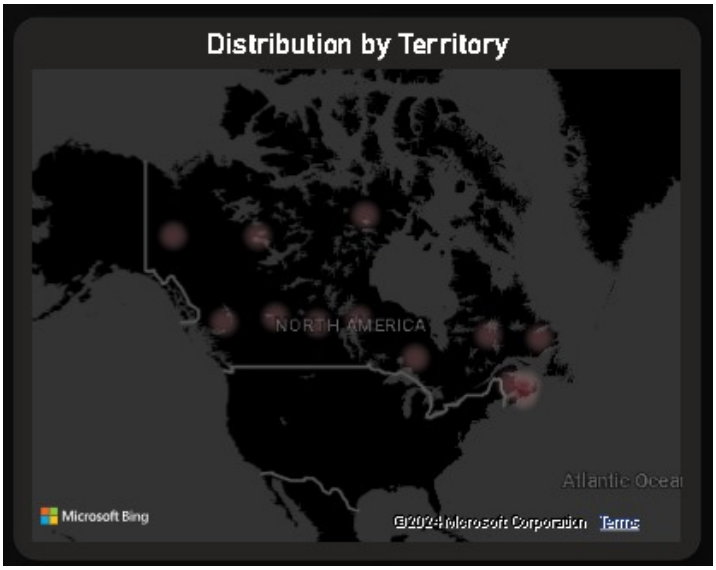
1.Line Chart

Relationship between each year quarter.

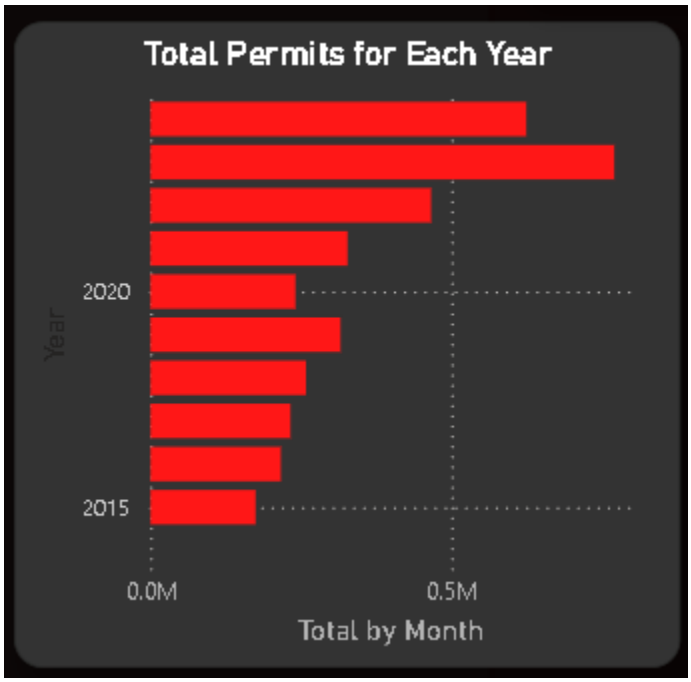
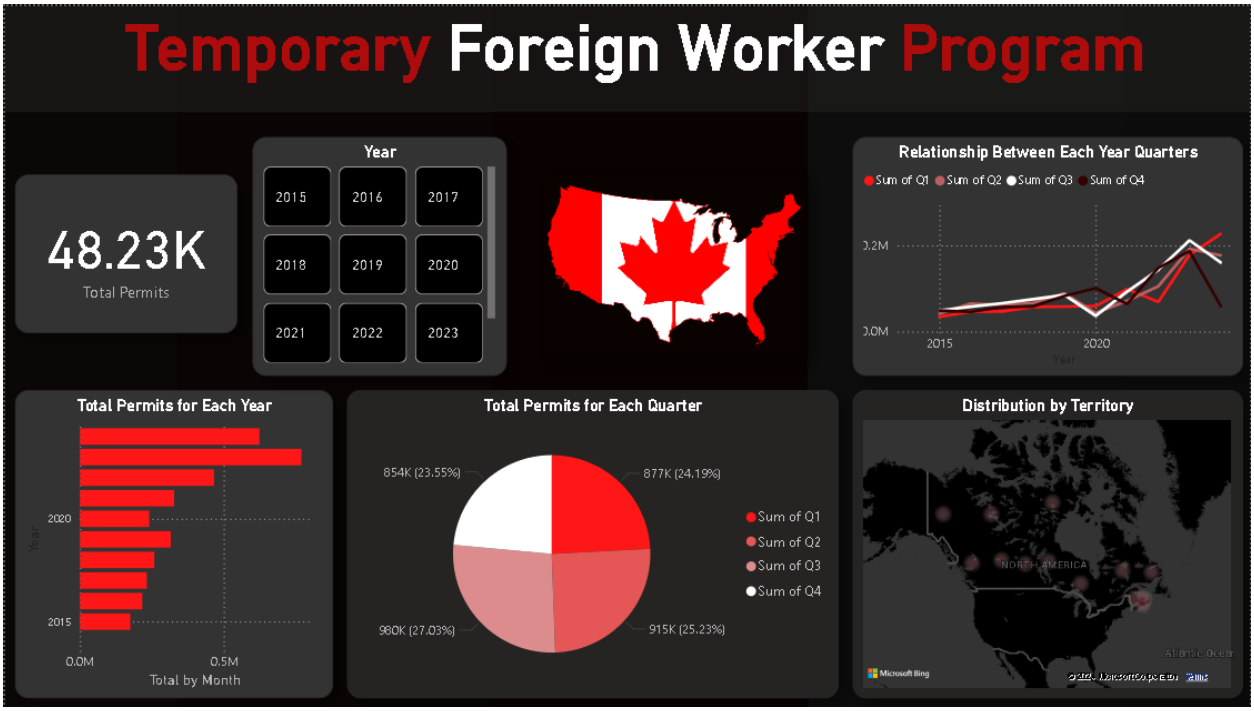


2.Heat map

Distribution by territory



3.Total permits for each year



4.Full Dashboard